Chapter III.7

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II.8.1. Here we will strengthen the results of the text to include information about the sheaf of differentials at a not necessarily closed point of a scheme X.

(a) Generalize (8.7) as follows. Let B be a local ring containing a field k, and assume that the residue field $k(B) = B/\mathfrak{m}$ of B is a separably generated extension of k. Then the exact sequence of (8.4A),

$$0 \to \mathfrak{m}/\mathfrak{m}^2 \xrightarrow{\delta} \Omega_{B/k} \otimes k(B) \to \Omega_{k(B)/k} \to 0$$

is exact on the left also. [*Hint*: In copying the proof of (8.7), first pass to $B/\mathfrak{m}^2$, which is a complete local ring, and then use (8.25A) to choose representatives for $B/\mathfrak{m}^2$.]

(b) Generalize (8.8) as follows. With B, k as above, assume furthermore that k is perfect, and that B is a localization of an algebra of finite type over k. Then show that B is a regular local ring if and only if $\Omega_{B/k}$ is free of rank $= \dim B +$ tr.d. $k(B)/k$.

(c) Strengthen (8.15) as follows. Let X be an irreducible scheme of finite type over a perfect field k, and let $\dim X = n$. For any point $x \in X$, not necessarily closed, show that the local ring $\mathscr{O}_{X,x}$ is a regular local ring if and only if the stalk $(\Omega_{X/k})_x$ of the sheaf of differentials at x is free of rank n.

(d) Strengthen (8.16) as follows. If X is a variety over and algebraically closed field k, then $U = \{x \in X | \mathscr{O}_x$ is a regular local ring$\}$ is an open dense subset of X.

II.8.6. *The Infinitesimal Lifting Prperty.*

6.10. *Duality for a Finite Flat Morphism.*

(a) Let $f : X \to Y$ be a finite morphism of noetherian schemes. For any quasi-coherent $\mathscr{O}_Y$-module $\mathscr{G}$, $\mathcal{H}om_Y(f_*\mathscr{O}_Y, \mathscr{G})$ is a quasi-coherent $f_*\mathscr{O}_Y$-module, and hence corresponds to a quasi-coherent $\mathscr{O}_X$-module, which we call $f^!\mathscr{G}$ (II, Ex. 5.17e).

(b) Show that for any coherent $\mathscr{F}$ on X and any quasi-coherent $\mathscr{G}$ on Y, there is a natural isomorphism

$$f_*\mathcal{H}om_X(\mathscr{F}, f^!\mathscr{G}) \xrightarrow{\cong} \mathcal{H}om_Y(f_*\mathscr{F}, \mathscr{G}).$$

(c) For each $i \geqslant 0$, there is a natural map

$$\phi_i : \operatorname{Ext}^i_X(\mathscr{F}, f^!\mathscr{G}) \to \operatorname{Ext}^i_Y(f_*\mathscr{F}, \mathscr{G})$$

[*Hint*: First construct a map

$$\operatorname{Ext}^i_X(\mathscr{F}, f^!\mathscr{G}) \to \operatorname{Ext}^i_Y(f_*\mathscr{F}, f_*f^!\mathscr{G}).$$

Then compose with a suitable map from $f_*f^!\mathscr{G} \to G$.]

(d) Now assume that X and Y are separated, $\mathfrak{Coh}(X)$ has enough locally frees, and assume that $f_*\mathscr{O}_X$ is locally free on Y (this is equivalent to saying f is flat–see Section 9). Show that ϕ_i is an isomorphism for all i, all $\mathscr{F}$ coherent on X, and all $\mathscr{G}$ quasi-coherent on Y. [*Hints*: First do $i = 0$. Then do $\mathscr{F} = \mathscr{O}_X$, using (Ex. 4.1). Then do $\mathscr{F}$ locally free. Do the general case by induction on i, writing $\mathscr{F}$ as a quotient of a locally free sheaf.]

Proof. □

7.2. Let $f : X \to Y$ be a finite morphism of projective schemes over a field k, and let ω_Y° be a dualizing sheaf for Y.
(a) Show that $f^!\omega_Y^\circ$ is a dualizing sheaf for X, where $f^!$ is defined as in (Ex. 6.10).
(b) If X and Y are both nonsingular, and k is algebraically closed, conclude that there is a natural trace map $t : f_*\omega_X \to \omega_Y$.

Proof. (a) Fix a coherent sheaf $\mathscr{F} \in \mathfrak{Mod}(X)$, and a trace map $t_Y : H^n(Y, \omega_Y^\circ) \to k$. (Ex. 6.10c) Gives us a map

$$\phi_0 : \mathrm{Hom}_X(\mathscr{F}, f^!\omega_Y^\circ) \xrightarrow{\cong} \mathrm{Hom}_Y(f_*\mathscr{F}, \omega_Y^\circ),$$

which is $f_*\mathscr{F}$ is coherent by (Ex. II.5.5), so there is a map

$$\mathrm{Hom}_Y(f_*\mathscr{F}, \omega_Y^\circ) \times H^n(Y, f_*\mathscr{F}) \to H^n(Y, \omega_Y^\circ)$$

which, composed with $t_Y : H^n(Y, \omega_Y^\circ) \to k$, induces a perfect pairing.
(b) Part (a), (Ex. 6.10b), and give us isomorphisms

$$\begin{aligned} f_*\omega_X &\cong f_*\mathcal{H}om_X(\mathscr{O}_X, \omega_X) \\ &\cong f_*\mathcal{H}om_X(\mathscr{O}_X, \omega_Y) \end{aligned}$$

□

7.3. Let $X = \mathscr{P}_k^n$. Show that $H^q(X, \Omega_X^p) = 0$ for $p \neq q$, k for $p = q, 0 \leqslant p, q \leqslant n$.

Proof. □

7.4. *The Cohomology Class of a Subvariety.* Let X be a nonsingular projective variety of dimension n over an algebraically closed field k. Let Y be a nonsingular subvariety of codimension p (hence dimension $n - p$). From the natural map $\Omega_X \otimes \mathscr{O}_Y \to \Omega_Y$ (II, 8.12) we deduce a map $\Omega_X^{n-p} \to \Omega_Y^{n-p}$. This induces a map on cohomology $H^{n-p}(X, \Omega_X^{n-p}) \to H^{n-p}(Y, \Omega_Y^{-n-p})$. Now $\Omega_Y^{n-p} = \omega_Y$ is a dualizing sheaf for Y, so we have the trace map $t_Y : H^{n-p}(Y, \Omega_Y^{n-p}) \to k$. Composing, we obtain a linear map $H^{n-p}(X, \Omega_X^{n-p}) \to k$. By (7.13) this corresponds to an element $\eta(Y) \in H^p(X, \Omega_X^p)$, which we call the *cohomology clas* of Y.
(a) If $P \in X$ is a closed point, show that $t_X(\eta(P)) = 1$, where $\eta(P) \in H^n(X, \Omega^n)$, and t_X is the trace map.
(b) If $X = \mathbb{P}^n$, identify $H^p(X, \Omega^p)$ with k by (Ex. 7.3), and show that $\eta(Y) = (\deg Y) \cdot 1$, where $\deg Y$ is its *degree* as a projective variety. [*Hint*: Cut with hyperplanes $H \subset X$, and use Bertini's Theorem (II, 8.18) to reduce to the case Y is a finite set of points.]

(c) For any scheme X of finite type over k, we define a homomorphism of sheaves of abelian groups $d\log : \mathscr{O}_X^\times \to \Omega_X$ by $d\log(f) = f^{-1}df$. Here $\mathscr{O}^\times$ is a group under multiplication, and Ω_X is a group under addition. This induces a map on cohomology $\operatorname{Pic} X = H^1(X, \mathscr{O}_X^\times) \to H^1(X, \Omega_X)$, which we denote by c-see (Ex. 4.5).

(d) Returning to the hypotheses above, suppose $p = 1$. Show that $\eta(Y) = c(\mathscr{L}(Y))$, where $\mathscr{L}(Y)$ is the invertible sheaf corresponding to the divisor Y.

Proof. (a) □